BrightLearn
MACHINE LEARNING

2026 EDITION · BEGINNER'S GUIDE

The Machine Learning Process

A complete beginner's guide to how machine learning models are built — from understanding a business problem all the way to monitoring a deployed model in the real world.

TOPIC End-to-End ML Workflow	**STEPS** 12 sequential steps
LEVEL Beginner-friendly	**INCLUDES** Definitions · Diagrams · Examples

Learn · Build · Deploy · Monitor
How real ML projects are built.

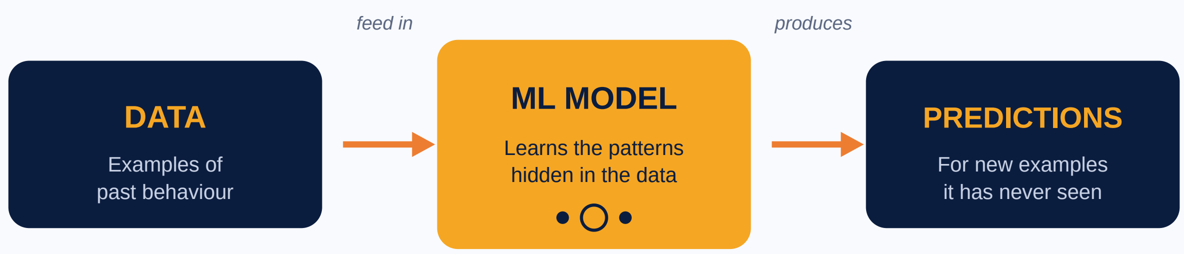
00 What is Machine Learning?

Before we walk through the 12-step process, let's make sure we're crystal clear on what machine learning actually *is*, and the key concepts you'll encounter throughout this guide.

DEFINITION

Machine Learning (ML)

Machine Learning is a way of teaching computers to **learn patterns from data** so they can make predictions or decisions **without being explicitly programmed for every possible case**. Instead of writing rules by hand, you give the computer many examples and let it figure out the rules itself.



The core idea of Machine Learning — patterns in, predictions out.

★ TRADITIONAL PROGRAMMING VS MACHINE LEARNING

In **traditional programming**, a developer writes the rules: "If a customer hasn't logged in for 30 days AND has fewer than 2 transactions, mark them as 'at risk'." The computer just follows the rules.

In **machine learning**, you give the computer thousands of past customers (some who left, some who stayed) and let it figure out the rules itself. It often discovers patterns a human would never have spotted.

01 Key Concepts You Need to Know

Three terms come up constantly in machine learning. Get comfortable with these now and the rest of this guide will make a lot more sense.

DEFINITION

Supervised Learning

Supervised learning is a type of machine learning where the model is taught using **labelled examples** — every example already has the correct answer attached. The model studies these answers and learns to predict the answer for new examples. **Most business ML problems** (churn, fraud, sales forecasts, credit scoring) are supervised learning.

FEATURES (inputs) + LABEL (known answer) → MODEL learns the link

Age	Spend	Complaints	Left Bank?
35	2500	1	No
42	800	4	Yes
29	3200	0	No

Each row already has the correct answer — that's what makes it "supervised."

In supervised learning, the training data includes the right answer alongside each set of inputs.

Within supervised learning, there are two main types of problems:

TYPE 1

Classification

Used when the answer you want to predict is a **category** — a yes/no, a label, a group. The output is one of a fixed set of options.

Examples: "Will this customer leave?" (Yes/No), "Is this email spam?" (Spam/Not spam), "Which product category?" (A/B/C).

TYPE 2

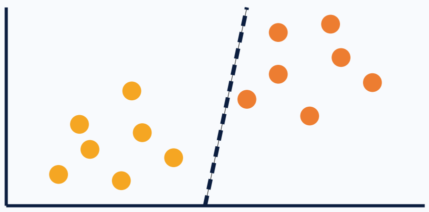
Regression

Used when the answer you want to predict is a **number** — a quantity, a price, a measurement. The output can be any value on a continuous scale.

Examples: "How much will this house sell for?" ($), "What will sales be next month?" (units), "How long until this part fails?" (days).

CLASSIFICATION

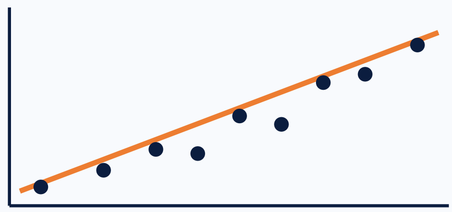
predicting categories



Stays Leaves

REGRESSION

predicting numbers



e.g. House size → predicted price

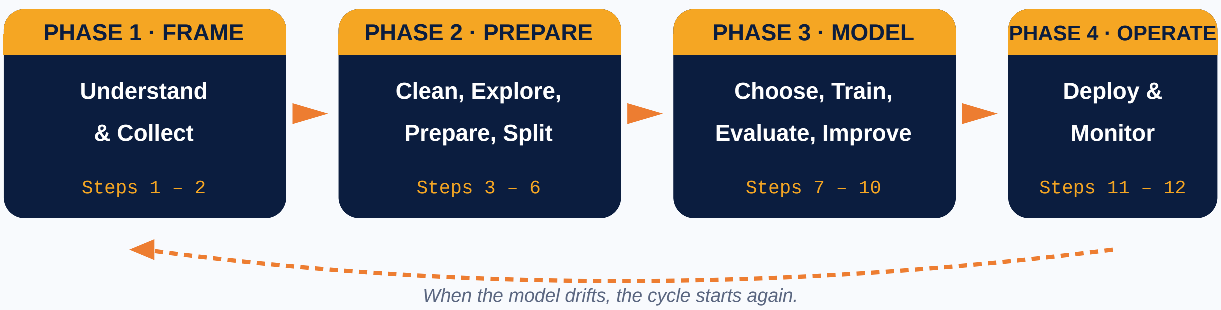
Classification draws a line between groups; Regression fits a line through points to predict a value.

02 The 12-Step ML Process

Every successful machine learning project — from Netflix recommendations to bank fraud detection — follows the same 12-step workflow. We've grouped the steps into four logical phases to make them easier to learn and remember.

★ THE FOUR PHASES AT A GLANCE

- **Phase 1 — Frame** (Steps 1–2): Understand the business problem and gather the data.
- **Phase 2 — Prepare** (Steps 3–6): Clean, explore, transform, and split the data.
- **Phase 3 — Model** (Steps 7–10): Choose an algorithm, train, evaluate, and improve.
- **Phase 4 — Operate** (Steps 11–12): Deploy the model and keep monitoring it over time.



The 12 steps fit into 4 phases — and the whole process is a loop, not a one-shot exercise.

★ THE MOST IMPORTANT THING TO REMEMBER

The single most common mistake in ML is **jumping straight to the algorithm**. Strong analysts spend most of their time on Phases 1 and 2 — understanding the problem and getting the data right. A great algorithm on bad data still gives bad answers. **Garbage in, garbage out.**

03 Phase 1 — Frame the Problem

Before writing a single line of code, you must understand *what you're trying to solve* and *what data exists to solve it*. This phase sets the direction for everything that follows.

PHASE 1 · FRAME

Steps 1 & 2 — Understand & Collect

01 Understand the Business Problem

Before building anything, you must understand the problem you are trying to solve. This step is about asking the right questions, not writing code.

EXAMPLE

A bank wants to predict whether a customer is likely to leave. The business problem could be:

> ***"How can we identify customers who may stop using our services?"***

Answering this single question determines what data you need, what type of model you'll build (in this case, a **classification** model), and how you'll measure success.

02 Collect Data

Machine learning needs data to learn patterns. Without enough relevant, good-quality data, no algorithm will give you useful answers.

EXAMPLE DATA

Customer ID	Age	Monthly Spend	Complaints	Left Bank
001	35	2500	1	No
002	42	800	4	Yes
003	29	3200	0	No

Data can come from **databases, Excel files, websites, apps, surveys, sensors,** or **APIs**. The more relevant historical examples you have, the better your model can learn.

⚠ COMMON PITFALL

Don't confuse "lots of data" with "good data." Ten thousand rows of incomplete or biased data will give you a worse model than one thousand rows of clean, representative data. Quality matters more than quantity.

04 Phase 2 — Prepare the Data

This is the longest and most important phase of any ML project. Cleaning, exploring, transforming, and splitting your data takes more time than building the model itself — but it's what makes the model trustworthy.

PHASE 2 · PREPARE

Steps 3 – 6 — Clean, Explore, Prepare, Split

03 Clean the Data

Raw data is almost always messy. Before training a model, you need to clean it up. This includes:

- Removing duplicate rows
- Fixing missing values
- Correcting wrong data
- Standardizing formats
- Removing unnecessary columns

EXAMPLE

If the gender column has values like:

```
Male, male, M, MALE
```

You may standardize them all to a single value:

```
Male
```

04 Explore and Understand the Data

This step is called **Exploratory Data Analysis**, or **EDA**. You analyze the data to understand patterns, trends, and relationships *before* you start modelling.

EXAMPLE QUESTIONS TO ASK

- Which customers are leaving?
- Does age affect customer churn?
- Do customers with many complaints leave more often?
- Which variables look most important?

You can use **charts, tables, averages,** and **summaries** to answer these questions. EDA often reveals surprises — like data quality issues you missed in Step 3, or relationships you didn't expect.

04 Phase 2 — Prepare the Data (continued)

Once your data is clean and you understand it, the next two steps reshape it into a form a machine learning algorithm can actually consume.

05 Prepare the Data for Machine Learning

Machine learning models need data in the right format. This step may include:

- Converting text into numbers
- Scaling numeric values
- Creating new useful columns (called **feature engineering**)
- Splitting the data into **features** and **target**

EXAMPLE

Features are the input columns the model uses to make a prediction:

```
Age, Monthly Spend, Complaints
```

Target is what you want the model to predict:

```
Left Bank
```

06 Split the Data

The data is usually split into two parts: one for the model to learn from, and one to test how well it learned.

Dataset	Purpose
`Training data`	Used to teach the model the patterns
`Testing data`	Used to check if the model actually works on new data

A typical 80 / 20 split

80% TRAINING	20% TEST
Model studies these examples	*Model never sees these*

Splitting forces the model to prove itself on data it has never seen — just like the real world.

⚠ COMMON PITFALL

Never test your model on the same data it learned from. If you do, the model will look perfect — because it has memorized the answers. The whole point of the test set is to be data the model has **never seen before**.

05 Phase 3 — Build the Model

Now the data is ready, you can choose the right algorithm, train the model, evaluate how well it does, and keep improving it. This is the part most people picture when they hear "machine learning" — but as you'll see, it's only a quarter of the work.

PHASE 3 · MODEL

Steps 7 – 10 — Choose, Train, Evaluate, Improve

07 Choose a Machine Learning Algorithm

An **algorithm** is the method the model uses to learn. Different algorithms are designed for different kinds of problems — picking the right one matters.

Algorithm	Used For
`Linear Regression`	Predicting numbers (regression)
`Logistic Regression`	Predicting yes/no outcomes (classification)
`Decision Tree`	Classification and prediction
`Random Forest`	More accurate decision-based models
`K-Means`	Grouping similar data
`Neural Networks`	Complex problems like images, speech, and text

EXAMPLE

To predict whether a customer will leave (a classification problem), you may try:

```
Logistic Regression
Decision Tree
Random Forest
```

08 Train the Model

Training means the model studies the training data and learns patterns. During training, the model adjusts itself thousands of times to make its predictions match the known answers as closely as possible.

EXAMPLE PATTERN THE MODEL MAY LEARN

> ***Customers with many complaints AND low monthly spending are more likely to leave.***

05 Phase 3 — Build the Model (continued)

Once trained, the model needs to be tested honestly — and almost always, improved. The best ML projects loop through Steps 8 – 10 several times before going live.

09 Evaluate the Model

After training, you test the model on the test data it has never seen before to measure how well it actually performs. Common evaluation metrics include:

Metric	Meaning
`Accuracy`	How many predictions were correct overall
`Precision`	How many predicted positives were actually correct
`Recall`	How many actual positives the model found
`F1 Score`	Balance between precision and recall
`RMSE`	Measures error for numeric predictions (regression)

EXAMPLE

If the model has **85% accuracy**, it means:

```
The model correctly predicted 85 out of 100 cases.
```

10 Improve the Model

The first model is rarely the best. You will almost always need to improve it. Ways to improve a model include:

- Add more data
- Clean the data better
- Remove bad columns
- Try different algorithms
- Tune model settings (called **hyperparameters**)
- Create better features

This step is repeated until the model performs well enough to deploy.

⚠ COMMON PITFALL

Beware of 100% accuracy. If your model is right every single time on the training data, it almost certainly hasn't learned the pattern — it has just memorized the examples. This is called **overfitting**, and it means your model will fail on real-world data. A slightly lower training accuracy with similar test accuracy is far healthier.

06 Phase 4 — Operate the Model

A model that lives only on your laptop creates no business value. The final phase is about putting your model into the real world — and watching it carefully once it gets there.

PHASE 4 · OPERATE

Steps 11 & 12 — Deploy & Monitor

11 Deploy the Model

Deployment means making the model available for real-world use. Until this step, the model is only useful to the people who built it.

EXAMPLE

A customer churn model can be added to a dashboard or application so the business can see things like:

> ***Customer 123 has a high risk of leaving.***

The model can be used in **websites, mobile apps, dashboards, APIs,** or **internal business systems** — anywhere a real decision needs to be made.

12 Monitor the Model

After deployment, you must keep checking the model. This is critical because **data can change over time** — a model that was excellent last year may quietly stop working this year.

EXAMPLE

Customer behavior today may not be the same as customer behavior next year. New competitors, new technology, and new economic conditions all change how people behave — a phenomenon called **data drift** or **model drift**.

Monitoring helps you know when to **retrain** or **update** the model.

★ THE ML PROCESS IS A LOOP, NOT A LINE

Once a model is monitored and starts to drift, you go back to earlier steps — collecting fresh data, re-cleaning, retraining, re-evaluating, and re-deploying. Real ML projects never truly "finish" — they get better, then they get retrained, then they get better again.

07 The Process at a Glance + A Worked Example

A one-page summary of all 12 steps in order, plus the same process applied to a different business problem so you can see the workflow holds for any ML project.

★ THE 12 STEPS IN ORDER

01 **Understand the Problem**
▼
02 **Collect Data**
▼
03 **Clean Data**
▼
04 **Explore Data (EDA)**
▼
05 **Prepare Data**
▼
06 **Split Data**
▼
07 **Choose Algorithm**
▼
08 **Train Model**
▼
09 **Evaluate Model**
▼
10 **Improve Model**
▼
11 **Deploy Model**
▼
12 **Monitor Model**

★ WORKED EXAMPLE — RETAIL

Scenario: A retail company wants to predict whether a customer will buy a product (a classification problem).

Step	What the team does
01	Frame the goal: "Predict customer purchases."
02	Gather customer transaction data.
03	Remove duplicates and fix missing values.
04	Analyze customer behavior and trends.
05	Convert text to numbers, scale, create features.
06	Split into 80% training / 20% testing.
07	Pick a classification algorithm (e.g. Logistic Regression).
08	Train the model on the training data.
09	Test the model on the test data.
10	Try other algorithms and tune settings.
11	Add it to the company's recommendation system.
12	Track predictions and retrain over time.

Patterns in.
Predictions out.

Machine learning is not magic — it's a careful 12-step process. Frame the problem, prepare the data, train and improve the model, then deploy and monitor it. Master this workflow and you can build ML solutions for any business in any industry.

★ BRIGHTLEARN MACHINE LEARNING ★